THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — SCIENCE · MEDIUM LEVEL CHAPTER TEST SET 2

Chapter 6: **How Forces Affect Motion** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. A ball of mass 0.2 kg moving at 5 m/s is caught and stopped in 0.05 s. The average force on the hand is: 1

(a) 200 N **(b)** 20 N **(c)** 1 N **(d)** 5 N

2. A horse pulls a cart with a force of 600 N, and the friction is 400 N. The cart's mass is 100 kg. The acceleration is: 1

(a) 10 m/s squared **(b)** 6 m/s squared **(c)** 2 m/s squared **(d)** 4 m/s squared

3. A boy of mass 40 kg runs at 5 m/s. His momentum is: 1

(a) 200 kg m/s **(b)** 35 kg m/s **(c)** 45 kg m/s **(d)** 8 kg m/s

4. A bullet of mass 20 g leaves a gun at 300 m/s. Its momentum is: 1

(a) 0.06 kg m/s **(b)** 600 kg m/s **(c)** 6 kg m/s **(d)** 60 kg m/s

5. The momentum of a body is 40 kg m/s and its velocity is 8 m/s. The mass is: 1

(a) 48 kg **(b)** 320 kg **(c)** 0.2 kg **(d)** 5 kg

6. Two bodies of masses 2 kg and 4 kg are acted on by the same force. The ratio of their accelerations is: 1

(a) 1 : 1 **(b)** 4 : 1 **(c)** 1 : 2 **(d)** 2 : 1

7. In free fall the apparent weight is: 1

(a) zero **(b)** m **(c)** 2mg **(d)** mg

8. A 10 kg trolley moving at 3 m/s is stopped by a constant force over 6 s. The force is: 1

(a) 0.5 N **(b)** 18 N **(c)** 5 N **(d)** 30 N

9. A net force of 10 N acts on a body of mass 5 kg for 6 s. The change in velocity is: 1

(a) 2 m/s **(b)** 12 m/s **(c)** 60 m/s **(d)** 30 m/s

10. A stone is thrown up. At the top of its path the net force on it is: 1

(a) its weight, downward **(b)** zero

(c) equal to its velocity **(d)** upward

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Explain why a gun recoils when it fires a bullet. 2

12. A ball of mass 0.5 kg hits a wall at 10 m/s and rebounds at 10 m/s. Find the change in momentum. 2

13. A bullet of mass 20 g is fired with 100 m/s. Find its momentum. 2

14. What is the net force on a body in free fall (ignoring air resistance) and its acceleration? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. A force of 30 N acts on a 5 kg body at rest on a smooth table. Find its acceleration and its speed after 4 s. 3

16. An object of mass 1 kg is dropped. Find its weight, its acceleration and the force with which the Earth is attracted by the object (g = 10). 3

17. State the relation between the mass of a body, its inertia and the force needed for the same acceleration, and compare a 2 kg and a 10 kg body for a = 1 m/s squared. 3

OR

Two forces of 6 N and 8 N act at right angles on a body of mass 2 kg. Find the net force and the acceleration. 3

18. Explain how a swimmer moves forward in water by Newton's third law and what role friction plays when we walk. 3

OR

Calculate the acceleration of a 70 kg skier on a slope when the downward pull along the slope is 210 N and friction is 70 N. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Explain the forces acting on a block pulled on a rough horizontal floor. Draw a free-body diagram in words and calculate the acceleration for m = 5 kg, pull = 30 N and friction = 10 N. 5

OR

Explain, using Newton's laws, the working of a hose pipe pushing a person who holds it and the recoil of a fire hose; calculate the force if 10 kg of water leaves per second at 6 m/s. 5

20. A car of mass 1000 kg starts from rest and reaches 20 m/s in 10 s. Find its acceleration, the net force and the distance travelled, and name the law used. 5

OR

Explain the law of inertia with the examples of a tablecloth pulled from under dishes, a coin on a card on a glass, and a vehicle turning a corner. 5

END OF QUESTION PAPER — **Total: 40 Marks**